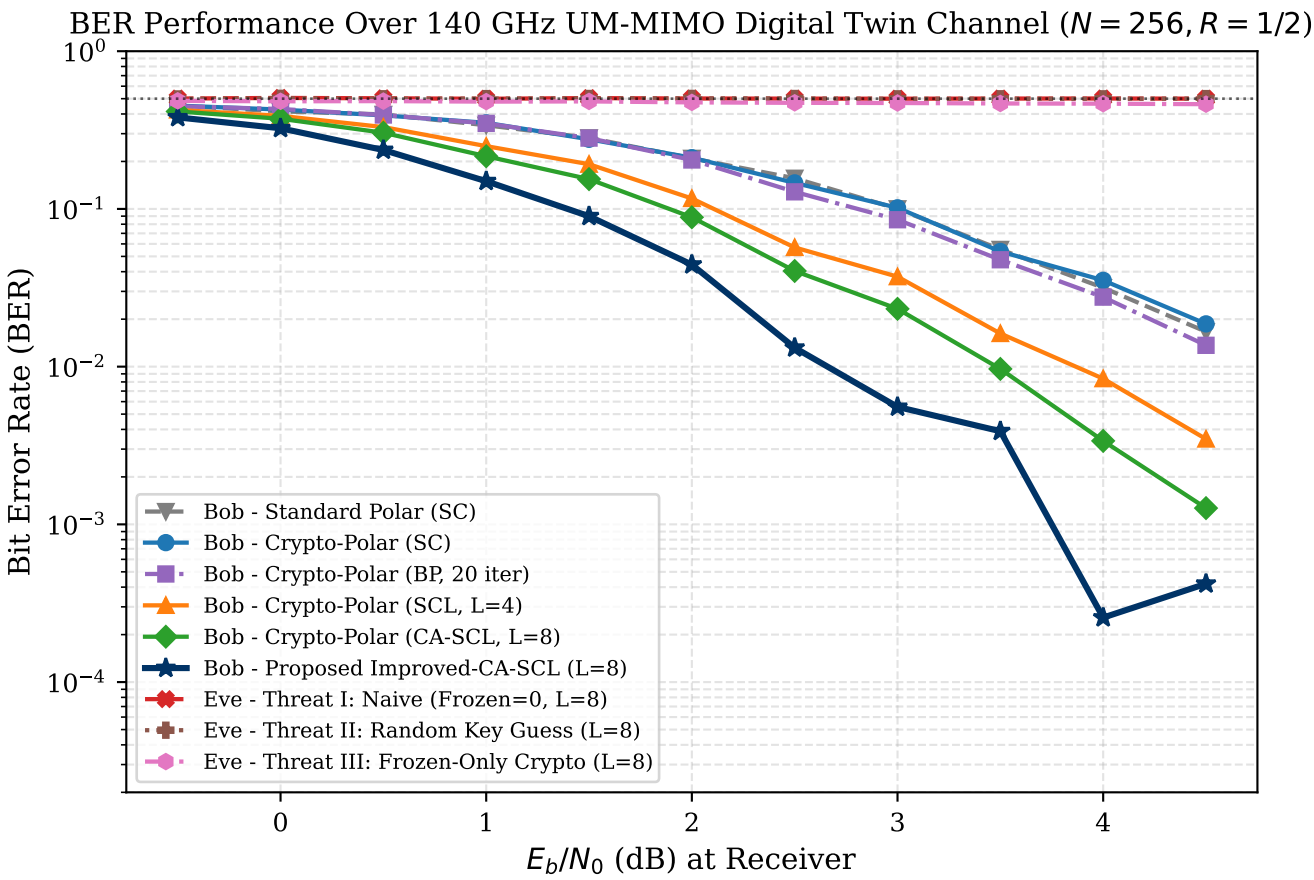
BER Performance Over 140 GHz UM-MIMO Digital Twin Channel ($N = 256, R = 1/2$)
Bit Error Rate (BER)
E_b/N_0 (dB) at Receiver
10^0
10^{-1}
10^{-2}
10^{-3}
10^{-4}
0
1
2
3
4
Bob - Standard Polar (SC)
Bob - Crypto-Polar (SC)
Bob - Crypto-Polar (BP, 20 iter)
Bob - Crypto-Polar (SCL, L=4)
Bob - Crypto-Polar (CA-SCL, L=8)
Bob - Proposed Improved-CA-SCL (L=8)
Eve - Threat I: Naive (Frozen=0, L=8)
Eve - Threat II: Random Key Guess (L=8)
Eve - Threat III: Frozen-Only Crypto (L=8)